

A Smooth Square-Root-Entropy Maximum for Attained Countable Affine Suprema

Candidate partial result for arXiv:1112.1042

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Status: candidate partial result, likely valid. The source question remains open for arbitrary convex functions on a separable infinite-dimensional Hilbert space.

1 Source question

Azagra asks whether every convex function on a separable infinite-dimensional Hilbert space which is bounded on bounded sets can be globally approximated by C^2 convex functions [1, p. 4]. The surrounding corollary and the exact question are reproduced below.

Conjecture. Let $U \subseteq \mathbb{R}^n$ be open and convex. For every convex function $f : U \rightarrow \mathbb{R}$ which is bounded on bounded subsets B of U with $\text{dist}(B, \partial U) > 0$, and for every $\varepsilon > 0$ there exists a C^1 convex function $g : U \rightarrow \mathbb{R}$ such that $f - \varepsilon \leq g \leq f$. Moreover g can be taken so as to preserve the set of minimizers and the local Lipschitz constants of f . And if f is strictly convex, so can g be taken.

A question remains open whether every convex function f defined on a separable infinite-dimensional Hilbert space X which is bounded on bounded sets can be globally approximated by C^2 convex functions (notice that Theorem 2 cannot be combined with the results of [8, 9] on smooth and real analytic approximation of bounded convex bodies in order to give a solution to this problem. Although one can use these results, together with the implicit function theorem, to find smooth convex approximations of f on a bounded set, the approximating functions obtained by this process are not defined on all of X and are not strongly convex, hence it is not clear how to extend them to a smooth convex function below f on X , or even if this should be possible at all).

Finally, as a byproduct of the proof of Theorem 1 we will also obtain the following characterization of the class of convex functions that can be globally approximated by strongly convex functions on $\mathbb{R}^n$.

Proposition 1. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. The following conditions are equivalent:*

- (1) *f cannot be uniformly approximated by strictly convex functions.*
- (2) *f cannot be uniformly approximated by strongly convex functions.*

Figure 1: The open question in arXiv:1112.1042, PDF page 4.

In the terminology of the paper, the desired one-sided uniform estimate is $f - \varepsilon \leq g \leq f$ on the whole space. The present packet proves this for a nontrivial class of globally Lipschitz convex functions.

2 Proof intuition

The usual countable log-sum-exp has a fatal global error: an affine function with a tiny reference weight pays a penalty proportional to the logarithm of that weight. We instead maximize over the probability simplex and add the bounded concave bonus

$$\delta \sum_i w_i \sqrt{\alpha_i}, \quad (w_i) \in \ell_2, \quad \sum_i \alpha_i = 1.$$

Cauchy–Schwarz bounds this bonus by $\delta \|w\|_2$, uniformly in the number of active affine functions. Its derivative is infinite at every coordinate face, so when the underlying countable maximum is attained, the unique maximizing probability has every coordinate positive.

The Lagrange multiplier λ is characterized by a single scalar equation. Attainment forces λ to lie strictly above the maximum. Uniformly bounded affine slopes then make the defining series smoothly convergent in a neighborhood of every point, and the Banach-space implicit function theorem gives a C^∞ multiplier and hence a C^∞ convex approximation. What remains open is how to replace the attained countable family by a smoothing device valid for an arbitrary dual domain.

3 The partial result

Let X be a real Banach space. Write an affine continuous function as

$$\ell_i(x) = x_i^*(x) + b_i, \quad x_i^* \in X^*, \quad b_i \in \mathbb{R}.$$

Theorem 1 (Square-root-entropy smoothing). *Suppose $(\ell_i)_{i \in \mathbb{N}}$ is a countable family of continuous affine functions on X such that*

$$L := \sup_i \|x_i^*\| < \infty,$$

and suppose

$$f(x) := \max_{i \in \mathbb{N}} \ell_i(x)$$

is finite and the displayed maximum is attained for every $x \in X$. Then, for every $\varepsilon > 0$, there is a C^∞ convex function $h : X \rightarrow \mathbb{R}$ such that

$$f - \varepsilon \leq h \leq f \quad \text{on } X.$$

More explicitly, fix positive weights w_i with $\sum_i w_i^2 = 1$, put $c = \varepsilon^2/4$, and let $\lambda(x) > f(x)$ be the unique solution of

$$c \sum_{i=1}^{\infty} \frac{w_i^2}{(\lambda(x) - \ell_i(x))^2} = 1. \tag{1}$$

Then one may take

$$h(x) = \lambda(x) + c \sum_{i=1}^{\infty} \frac{w_i^2}{\lambda(x) - \ell_i(x)} - \varepsilon. \tag{2}$$

Proof. Let

$$\Delta = \left\{ \alpha = (\alpha_i) \in \ell_1 : \alpha_i \geq 0, \sum_i \alpha_i = 1 \right\}.$$

Consider the regularized maximum

$$g(x) = \sup_{\alpha \in \Delta} \left\{ \sum_i \alpha_i \ell_i(x) + \varepsilon \sum_i w_i \sqrt{\alpha_i} \right\}. \tag{3}$$

The affine average is interpreted in the extended sense

$$\sum_i \alpha_i \ell_i(x) := f(x) + \sum_i \alpha_i (\ell_i(x) - f(x)) \in [-\infty, f(x)].$$

This is unambiguous because every term in the last series is nonpositive. (Equivalently, the bounded-slope hypothesis makes the linear part absolutely convergent, while a divergent intercept part can only have value $-\infty$.) For each $\alpha \in \Delta$, Cauchy–Schwarz gives

$$\sum_i w_i \sqrt{\alpha_i} \leq \left(\sum_i w_i^2 \right)^{1/2} \left(\sum_i \alpha_i \right)^{1/2} = 1.$$

Moreover, $\sum_i \alpha_i \ell_i(x) \leq f(x)$. Conversely, concentrating all mass at one index and taking the supremum over that index yields $g(x) \geq f(x)$. Thus

$$f(x) \leq g(x) \leq f(x) + \varepsilon. \quad (4)$$

As a supremum of affine functions of x , with an x -independent bonus, g is convex.

Fix $x \in X$ and set $M = f(x)$. For $s > M$, define

$$\Phi_x(s) = c \sum_i \frac{w_i^2}{(s - \ell_i(x))^2}.$$

This is finite, continuous, and strictly decreasing on (M, ∞) , and it tends to zero as $s \rightarrow \infty$. Since the maximum defining M is attained, say $\ell_k(x) = M$, the k th summand shows that $\Phi_x(s) \rightarrow \infty$ as $s \downarrow M$. Hence there is a unique $\lambda(x) > M$ satisfying (1).

Define

$$\alpha_i(x) = c \frac{w_i^2}{(\lambda(x) - \ell_i(x))^2}. \quad (5)$$

Equation (1) says that $\alpha(x) \in \Delta$, and every coordinate is positive. It is the unique optimizer in (3). Indeed, concavity of the square root gives, for every $a \geq 0$ and every $r > 0$,

$$\sqrt{a} \leq \sqrt{r} + \frac{a - r}{2\sqrt{r}}.$$

Apply this with $r = \alpha_i(x)$. From (5),

$$\frac{\varepsilon w_i}{2\sqrt{\alpha_i(x)}} = \lambda(x) - \ell_i(x).$$

Put $d_i = \lambda(x) - \ell_i(x) > 0$. The tangent inequality says coordinatewise that

$$-d_i \alpha_i + \varepsilon w_i \sqrt{\alpha_i} \leq -d_i \alpha_i(x) + \varepsilon w_i \sqrt{\alpha_i(x)} = c \frac{w_i^2}{d_i}.$$

The right-hand series converges because $d_i \geq \lambda(x) - f(x) > 0$. Consequently the series on the left has a summable positive part and is well-defined in $[-\infty, \infty)$. Summing the inequalities and using $\sum_i \alpha_i = \sum_i \alpha_i(x) = 1$ proves that the objective at α is at most the objective at $\alpha(x)$. If equality holds, the strict tangent inequality for $\sqrt{\cdot}$ forces every $\alpha_i = \alpha_i(x)$, proving uniqueness. At the optimizer the affine average is absolutely convergent: with $d_i = \lambda(x) - \ell_i(x)$,

$$\sum_i \alpha_i(x) |\ell_i(x)| \leq |\lambda(x)| + c \sum_i \frac{w_i^2}{d_i} < \infty.$$

Using $d_i = \lambda(x) - \ell_i(x)$ and $\sum_i \alpha_i(x) = 1$, the optimizer value simplifies to

$$\begin{aligned} g(x) &= \sum_i \frac{cw_i^2}{d_i^2} \ell_i(x) + \sum_i \frac{\varepsilon^2 w_i^2}{2d_i} \\ &= \lambda(x) + c \sum_i \frac{w_i^2}{d_i}. \end{aligned}$$

This is the expression in (2) before subtracting ε .

It remains to prove smoothness. Set

$$F(x, s) = c \sum_i \frac{w_i^2}{(s - \ell_i(x))^2} - 1 \quad (s > f(x)).$$

Fix (x_0, s_0) with $s_0 = \lambda(x_0)$ and put $\eta = s_0 - f(x_0) > 0$. Since $\sup_i \|x_i^*\| = L$, after restricting to a small product neighborhood of (x_0, s_0) all denominators are at least $\eta/2$. Every derivative of every summand is bounded there by a constant depending only on its order, η , and L , times w_i^2 . The series and all of its derivatives therefore converge uniformly on that neighborhood. Thus F is C^∞ , and

$$\partial_s F(x_0, s_0) = -2c \sum_i \frac{w_i^2}{(s_0 - \ell_i(x_0))^3} < 0.$$

The implicit function theorem on Banach spaces shows that λ is C^∞ near x_0 . The same locally uniform derivative estimates show that the series in the formula for g is C^∞ . Since x_0 was arbitrary, g is C^∞ on X .

Finally let $h = g - \varepsilon$. Convexity and smoothness are unchanged, and (4) becomes $f - \varepsilon \leq h \leq f$. $\square$

Corollary 2. *On ℓ_2 , the convex Lipschitz function*

$$f(x) = \max\{0, \sup_{n \geq 1} x_n\}$$

admits global C^∞ convex minorants within every prescribed uniform error.

Proof. Use the affine family $0, x_1, x_2, \dots$. Its slopes have norm at most one. Since $x_n \rightarrow 0$ for every $x \in \ell_2$, a positive supremum is attained by one of the coordinates, while a nonpositive supremum is attained by the zero affine function. Theorem 1 applies. $\square$

This example is the support function of the closed convex hull of $\{0, e_1, e_2, \dots\}$. It is precisely the infinite-simplex stress test on which summable-weight log-sum-exp fails: square-root entropy removes the unbounded logarithmic penalty.

4 Limitations and deep-upgrade audit

The theorem does not solve the source question. An arbitrary Lipschitz convex function on an infinite-dimensional Hilbert space need not be an attained countable maximum of affine functions. For example, a countable dense family of linear supports represents the Hilbert norm as a supremum, but the supremum is generally not attained. At a nonattainment point the left side of (1) need not diverge as $s \downarrow f(x)$, so the strict multiplier gap—and hence the smooth implicit formula—can fail.

A further upgrade attempt asked whether every Lipschitz convex function can first be approximated uniformly by an attained countable affine maximum. For support functions, uniform

additive approximation fixes the recession function. In infinite dimension, forcing attainment with a countable set of slopes amounts to a compactness/properness property unavailable on the whole unit sphere; simple decaying penalties again allow minimizing sequences to escape. Thus this route does not reduce the full question to the theorem.

The earlier standard routes—Moreau regularization, finite-rank convolution, Gaussian convolution, quadratic convexity repair, and countable log-sum-exp—are documented for the stronger later question in arXiv:1411.0471 [2]. Moreau regularization stops at $C^{1,1}$, while a nondegenerate Gaussian covariance cannot be trace class in infinite dimension. The present result is a genuine improvement over the countable-max route, but not over the arbitrary-function problem.

5 Novelty check and verification

A bounded search used the source id and title; the exact phrases “separable infinite-dimensional Hilbert space” and “ C^2 convex functions”; the later paper arXiv:1411.0471; and combinations of “countable maximum”, “square-root entropy”, “Tsallis entropy”, “smooth maximum”, and “affine functions”. It found the source papers and unrelated uses of Tsallis entropy, but no statement of Theorem 1. This supports only bounded novelty confidence.

The proof was checked algebraically in four places: the uniform entropy bound, the scalar multiplier equation, the tangent inequality proving the optimizer, and local uniform convergence of every differentiated series. No numerical computation is used. Human review should focus on the infinite-simplex optimization and the passage from the scalar implicit equation to Fréchet C^∞ smoothness.

References

- [1] D. Azagra, *Global approximation of convex functions*, arXiv:1112.1042v7 (2012).
- [2] D. Azagra and C. Mudarra, *Global approximation of convex functions by differentiable convex functions on Banach spaces*, arXiv:1411.0471 (2014).